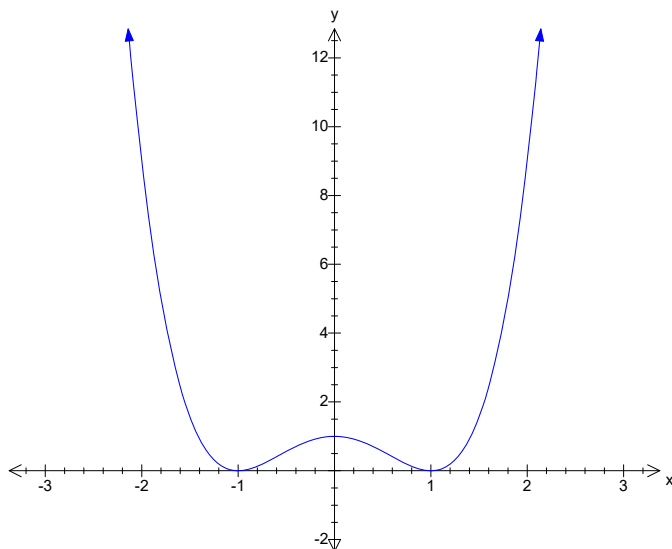


QUESTION 1 (10 MARKS) Start this question on a new page.

(a) Simplify $\frac{1}{\sqrt{3}+2} - \frac{1}{\sqrt{3}-2}$. 2

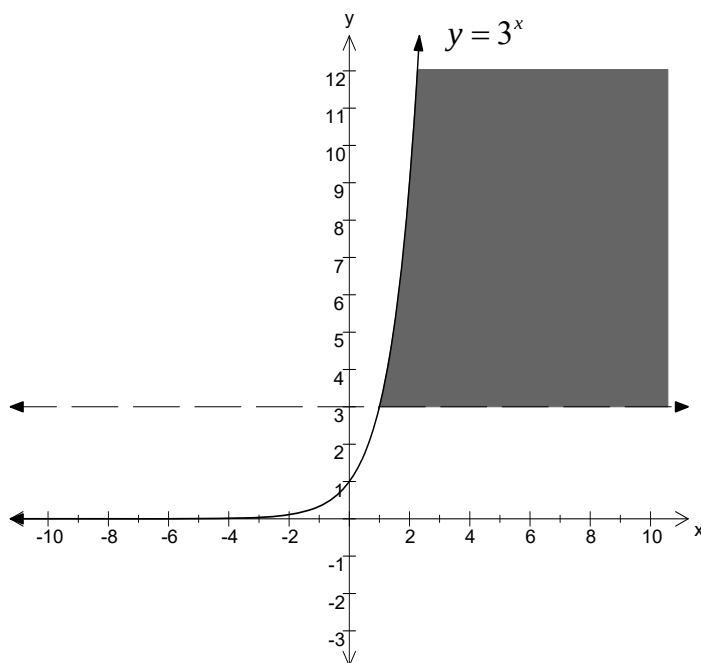
(b) Copy the graph of $f(x)$ below on to your answer paper. On the same axes sketch the graph of $f(x-1)$. 1



(c) If $g(x) = \frac{x}{x-3}$ find the inverse function $g^{-1}(x)$. 2

(d) Solve the inequation $\frac{1}{x-4} \geq 2$. 3

(e) Define the shaded region below 2

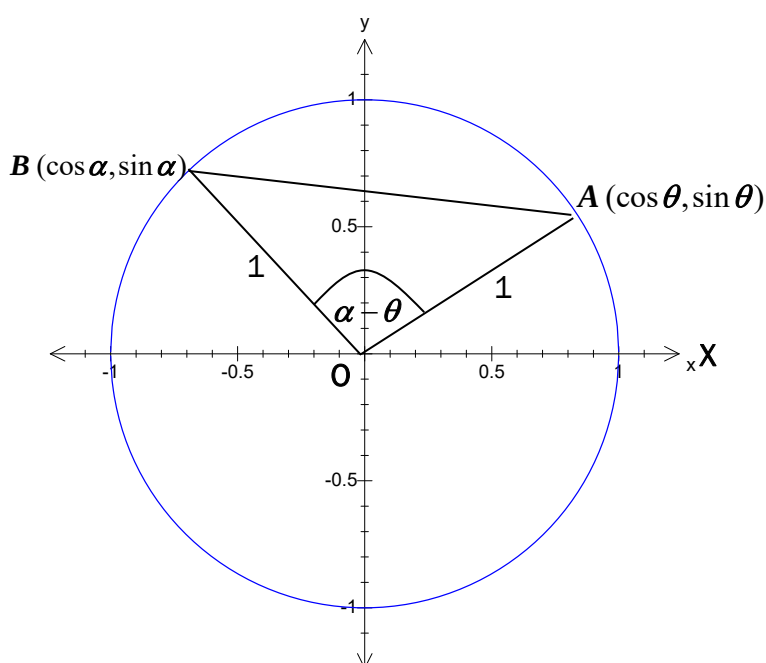


QUESTION 2 (10 MARKS) Start this question on a new page.

(a) Solve $\tan 2\theta = \sqrt{3}$ where $0 \leq \theta \leq 360^\circ$. **3**

(b) Prove the identity $\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \operatorname{cosec}^2 \theta} = 1$. **2**

(c) Consider the unit circle below. Let $\angle XO\hat{A} = \theta$ and $\angle XO\hat{B} = \alpha$. **5**



(i) By considering the triangle OAB, show that $\cos(\alpha - \theta) = \frac{2 - AB^2}{2}$

(ii) Using the distance formula, show that
 $AB^2 = 2 - 2 \cos \alpha \cos \theta - 2 \sin \alpha \sin \theta$

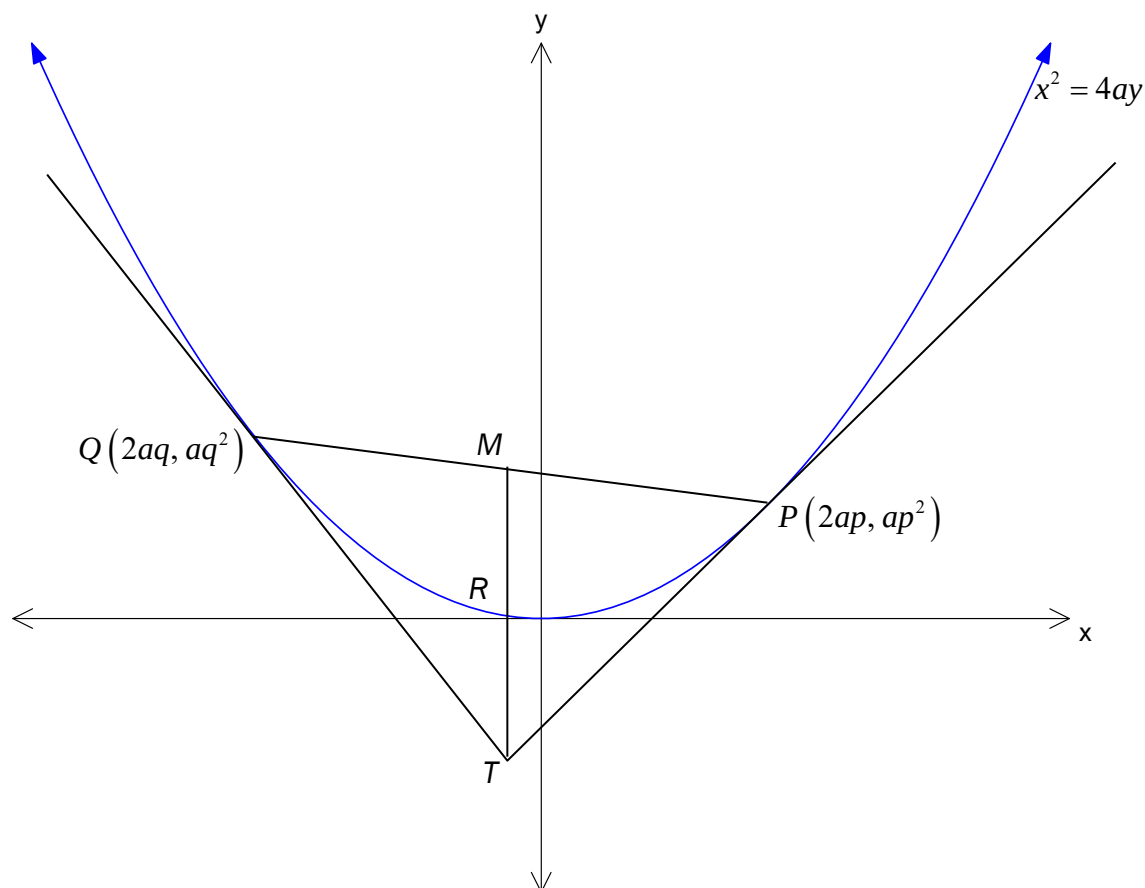
(iii) Hence show that $\cos(\alpha - \theta) = \cos \alpha \cos \theta + \sin \alpha \sin \theta$

QUESTION 3 (10 MARKS) Start this question on a new page.

- (a) The fourth term of an arithmetic sequence is 30 and the sixth term is 42. **3**
- (i) Find the 12^{th} term of the sequence.
- (ii) Find the sum of the first 10 terms.
- (b) Evaluate $\sum_{r=1}^5 (-1)^r 2r$ **1**
- (c) The sum of the first n terms of a sequence is given by $S_n = 32 - 3n^2$. **3**
- (i) Show that the n^{th} term of this sequence is given by $T_n = 3 - 6n$.
- (ii) Find the value of the first term of the sequence that is less than -100 .
- (d) Prove by induction that $5^n + 3$ is divisible by 4 for any positive integer n . **3**

QUESTION 4 (12 MARKS) Start this question on a new page.

- (a) If $x = t + 1$ and $y = \frac{1}{t - 2}$, eliminate the parameter t to find the Cartesian equation. 2
- (b) Consider the parabola $(y - 1)^2 = 8(x + 2)$. Find 2
- the coordinates of the vertex
 - the coordinates of the focus
- (c) P and Q are two points on the parabola $4ay = x^2$ with parameters p and q respectively. The tangents at P and Q meet at T . M is the midpoint of PQ . 8
- Find the equation of the tangent at P and hence state the equation of the tangent at Q .
 - Show that these tangents meet at the point $T(a(p + q), apq)$.
 - Find the midpoint M of the interval PQ .
 - Show that TM is parallel to the axis of the parabola.
 - Show that the point R where TM intersects the parabola is the midpoint of TM .



QUESTION 5 (12 MARKS) Start this question on a new page.

(a) If $f(x) = (2x+1)(x^3-1)^2$ find $f'(-1)$. **2**

(b) Using the definition of the derivative $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ **2**
find $f'(x)$ where $f(x) = x - 2x^2$.

(c) Consider the function $y = \frac{x^2}{\sqrt{2x+1}}$. **5**

(i) State the domain of the function

(ii) Show that $\frac{dy}{dx} = \frac{x(3x+2)}{\sqrt{(2x+1)^3}}$

(iii) Hence find the point or points on the curve where the function is horizontal.

(d) An equilateral triangle has sides of length s cm and area A cm². **3**

(i) Show that $A = \frac{\sqrt{3}s^2}{4}$.

(ii) If the sides are increasing at a constant rate of 2 cm/second, find the rate at which the area is increasing when the sides are of length 3 cm.

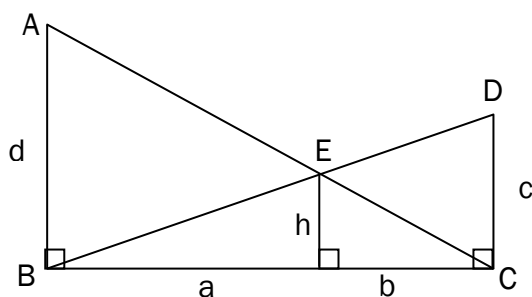
QUESTION 6 (9 MARKS) Start this question on a new page.

- (a) Solve $x^6 - 7x^3 - 8 = 0$. 2
- (b) Express $2x^2 - 7x - 4$ in the form $a(x+2)^2 + b(x+2) + c$. 3
- (c) Find the values of m if the equation $x^2 - 3mx + (m+3) = 0$ has: 4
- (i) One root double the other (hint: let the roots be α and 2α)
- (ii) One root the reciprocal of the other

QUESTION 7 (10 MARKS) Start this question on a new page.

- (a) If A is the point (10, 2) and B is the point (-2, 6), find the point P dividing AB externally in the ratio 4:3. 2
- (b) The perpendicular distance between the origin and the line $x - 2y + k = 0$ is $\sqrt{5}$. Find two possible values of k . 3

- (c) 5



- (i) Show that $\frac{h}{d} = \frac{b}{a+b}$, stating all reasons.
- (ii) By stating a similar expression for $\frac{h}{c}$, show that $\frac{1}{c} + \frac{1}{d} = \frac{1}{h}$.

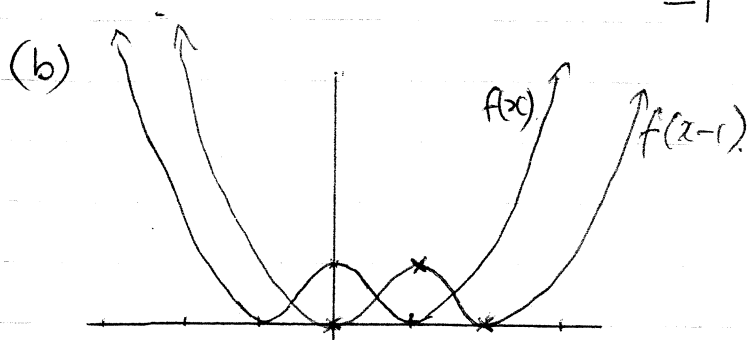
END OF PAPER

Y11 Ext 1 Prelim 2008

Question 1

$$(a) \frac{1}{\sqrt{3}+2} - \frac{1}{\sqrt{3}-2} = \frac{(\sqrt{3}-2) - (\sqrt{3}+2)}{(\sqrt{3}+2)(\sqrt{3}-2)}$$

$$= \frac{-4}{-1} = 4.$$



(c). let $y = \frac{x}{x-3}$ for inverse swap x, y

a $x = \frac{y}{y-3}$

$$(y-3)x = y$$

$$xy - 3x = y$$

$$xy - y = 3x$$

$$y(x-1) = 3x$$

$$y = \frac{3x}{x-1}$$

$$g^{-1}(x) = \frac{3x}{x-1}$$

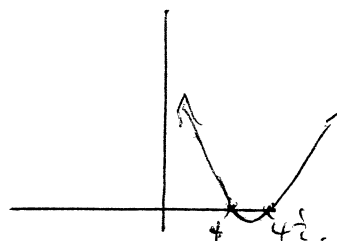
(d) $x(x-4)^2 - (x-4) \geq 2(x-4)^2 \quad (x \neq 4)$

$$2(x-4)^2 - (x-4) \leq 0$$

$$(x-4)[2(x-4) - 1] \leq 0$$

$$(x-4)(2x-9) \leq 0$$

$$4 < x \leq \frac{9}{2}$$

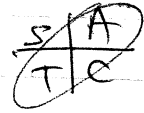


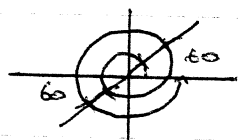
(e). $y > 3, y \leq 3^x$

QUESTION 2

(a) domain $0 \leq \theta \leq 360$
 $0 \leq 2\theta \leq 720$.

base $< 60^\circ$

Quadrants 1, 3. 



$2\theta = 60, 240, 420, 600$

$\theta = 30, 120, 210, 300$.

(b) LHS = $\frac{1}{1+\sin^2\theta} + \frac{1}{1+\cos^2\theta}$
 $= \frac{1}{1+\sin^2\theta} + \frac{1}{1+\frac{1}{\sin^2\theta}}$
 $= \frac{1}{1+\sin^2\theta} + \frac{\sin^2\theta}{\sin^2\theta+1}$
 $= \frac{1+\sin^2\theta}{1+\sin^2\theta} = 1 = \text{RHS.}$

(c)(i) $\cos(x-\theta) = \frac{l^2 + l^2 - AB^2}{2 \times l \times l}$
 $= \frac{2 - AB^2}{2}$

(ii) $AB^2 = (\cos\theta - \cos x)^2 + (\sin\theta - \sin x)^2$
 $= \cos^2\theta - 2\cos\theta\cos x + \cos^2 x + \sin^2\theta - 2\sin\theta\sin x + \sin^2 x$
 $= (\sin^2\theta + \cos^2\theta) + (\sin^2 x + \cos^2 x) - 2\cos x \cos \theta - 2\sin x \sin \theta$
 $= 2 - 2\cos x \cos \theta - 2\sin x \sin \theta$ (since $\sin^2 x + \cos^2 x = 1$)

(iii) From (i) and (ii)

$\cos(x-\theta) = \frac{2 - [2 - 2\cos x \cos \theta - 2\sin x \sin \theta]}{2}$
 $= \frac{2\cos x \cos \theta + 2\sin x \sin \theta}{2}$
 $= \cos x \cos \theta + \sin x \sin \theta$

QUESTION 3

$$(a) T_4 = a + 3d = 30$$

$$T_6 = a + 5d = 42$$

$$2d = 12$$

$$d = 6$$

$$a = 12$$

$$(i) T_{12} = \dots + 11 \times \dots =$$

$$(ii) S_{10} = \frac{10}{2} (2 \times 12 + 11 \times 6)$$

$$= 450$$

$$(b) \sum_{r=1}^5 (-1)^r 2r = -2 + 4 - 6 + 8 - 10$$

$$= -6$$

$$(c)(i) T_n = S_n - S_{n-1}$$

$$= 32 - 3n^2 - (32 - 3(n-1)^2)$$

$$= 32 - 3n^2 - 32 + 3(n^2 - 2n + 1)$$

$$= 32 - 3n^2 - 32 + 3n^2 - 6n + 3$$

$$= 3 - 6n$$

$$(ii) \text{ find } n \text{ such that } T_n < -100$$

$$\text{i.e. } 3 - 6n < -100$$

$$-6n < -103$$

$$n > \frac{103}{6} \left(17\frac{1}{6} \right)$$

$$\text{thus first term } < -100 \text{ is } T_{18} = 3 - 6 \times 18$$

$$= -105$$

$$(d) \text{ Prove } 5^n + 3 = 4p \text{ (p integer)}$$

Show true for $n=1$

$$\text{LHS} = 5^1 + 3$$

$$= 8$$

$$= 4 \times 2 \text{ is divisible by 4.}$$

Suppose true for $n=k$

$$\text{i.e. } 5^k + 3 = 4m \text{ (m integer)}$$

$$5^k = 4m - 3 \quad (*)$$

Show true for $n=k+1$

$$\text{i.e. } 5^{k+1} + 3 = 4q \text{ (q integer)}$$

$$\text{LHS} = 5 \times 5^k + 3$$

$$= 5 \times (4m - 3) + 3 \text{ from assumption } (*)$$

$$= 20m - 15 + 3$$

$$= 20m - 12$$

$$= 4(5m - 3)$$

$$= 4q \quad (q = 5m - 3)$$

Hence by mathematical induction true for all $n \geq 1$.

QUESTION 4.

(a) $x = t + 1$

then $t = x - 1$.

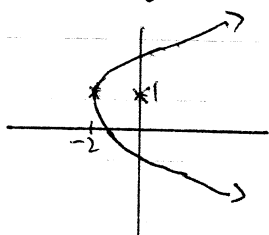
substitute for t in $y = \frac{1}{t-2}$.

$$y = \frac{1}{(x-1)-2}$$

$$y = \frac{1}{x-3}$$

(b) (i) vertex is $(-2, 1)$

(ii) focal length is 2.



focus is $(0, 1)$

(c) (i) $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

At $x = p$, $\frac{dy}{dx} = p$.

Equation of tangent at P is $y - ap^2 = p(x - 2ap)$

$$y = px - ap^2 \quad (*)$$

Hence tangent at Q is $y = qx - aq^2$

(ii) these meet when $px - ap^2 = qx - aq^2$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p + q)(p - q)$$

$$x = a(p + q)$$

sub in (*) $y = p \cdot a(p + q) - ap^2$

$$y = apq$$

Hence point of intersection is T $(a(p+q), apq)$

(iii) Midpoint of PQ $M = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$

$$= \left(a(p+q), \frac{a(p^2 + q^2)}{2} \right)$$

(iv) Since x -value of M and T are the same equation of MT is $x = a(p+q)$ is parallel to the axis of the parabola. $= 0$

(v) Midpoint of MT is $R = \left(a(p+q), \frac{2apq + ap^2 + aq^2}{4} \right)$
 $= \left(a(p+q), \frac{a(p+q)^2}{4} \right)$.

If R lies on parabola it must satisfy $x^2 = 4ay$.

$$LHS = a^2(p+q)^2$$

$$RHS = \frac{4a \times a(p+q)^2}{4}$$

$$= a^2(p+q)^2$$

Since $LHS = RHS$ R lies on MT .

QUESTIONS

(a) $f(x) = (2x+1)(x^3-1)^2$

$f'(x) = (2x+1) 2(x^3-1) 3x^2 + (x^3-1)^2 \times 2$ ✓

$f'(-1) = -1 \times 2 \times -2 \times 3 \times 1 + (-2)^2 \times 2$

$= 20$ ✓ (2)

(b) $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) - 2(x+h)^2 - (x - 2x^2)}{h}$ ✓

$= \lim_{h \rightarrow 0} \frac{x+h - 2(x^2 + 2xh + h^2) - x + 2x^2}{h}$

$= \lim_{h \rightarrow 0} \frac{\cancel{x} + h - \cancel{2x^2} - 4xh - 2h^2 - \cancel{x} + \cancel{2x^2}}{h}$

$= \lim_{h \rightarrow 0} \frac{\cancel{h}(1 - 4x - 2h)}{\cancel{h}}$ ✓

$= 1 - 4x$ (2)

(c) (i) domain is $x > -\frac{1}{2}$ ✓

(ii) $y = \frac{x^2}{(2x+1)^{\frac{1}{2}}}$

$\frac{dy}{dx} = \frac{(2x+1)^{\frac{1}{2}} \times 2x - x^2 \times \frac{1}{2} (2x+1)^{-\frac{1}{2}}}{(2x+1)}$ ✓

$\times \frac{(2x+1)^{\frac{1}{2}}}{(2x+1)^{\frac{1}{2}}}$

$= \frac{(2x+1)2x - x^2}{(2x+1)^{\frac{3}{2}}}$

$= \frac{4x^2 + 2x - x^2}{(2x+1)^{\frac{3}{2}}}$

$= \frac{3x^2 + 2x}{(2x+1)^{\frac{3}{2}}}$ ✓

$= \frac{x(3x+2)}{(2x+1)^{\frac{3}{2}}}$ as required

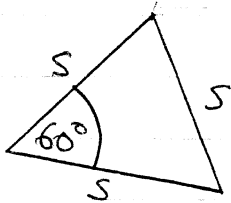
(iii) curve is horizontal when $\frac{dy}{dx} = 0$

i.e. $x(3x+2) = 0$ ✓

$x = 0$ $x = -\frac{2}{3}$ ✓

but since domain is $x > -\frac{1}{2}$ $x = 0$ ✓ is only solution.

(d)(i)



$$\begin{aligned}\text{Area} &= \frac{1}{2} \times s \times s \times \sin 60 \\ &= \frac{1}{2} s^2 \times \frac{\sqrt{3}}{2} \\ &= \frac{\sqrt{3} s^2}{4}\end{aligned}$$

$$(ii) \quad A = \frac{\sqrt{3} s^2}{4} \quad \frac{dA}{ds} = \frac{\sqrt{3} s}{2}$$

$$\frac{dA}{dt} = \frac{dA}{ds} \times \frac{ds}{dt}$$

$$\begin{aligned}\frac{dA}{dt} &= \frac{\sqrt{3} s}{2} \times 2 \\ &= \sqrt{3} s.\end{aligned}$$

When $s = 3$.

$$\frac{dA}{dt} = 3\sqrt{3} \text{ cm}^2/\text{sec}.$$

(3)

QUESTION 6

(a) $(x^3)^2 - 7x^3 - 8 = 0$
 $(x^3 + 1)(x^3 - 8) = 0$
 $x^3 = -1 \quad x^3 = 8$
 $x = -1, 2$

(b) $2x^2 - 7x - 4 \equiv a(x+2)^2 + b(x+2) + c$

Equating coefficients of x^2 $a = 2$

Substituting $x = -2$ $c = \dots$

substituting $x = 0$ $-4 = 4a + 2b + c$

$$-4 = 8 + 2b + 18$$

$$2b = -30$$

$$b = -15$$

$$\therefore 2x^2 - 7x - 4 \equiv 2(x+2)^2 - 15(x+2) + 18$$

(c) (i) let roots be x and $2x$.

then $x + 2x = 3m$

$$3x = 3m$$

$$x = m \quad *$$

and $x \times 2x = m + 3$

$$2x^2 = m + 3$$

from $*$ $2m^2 = m + 3$

$$2m^2 - m - 3 = 0$$

$$(2m - 3)(m + 1) = 0$$

$$m = \frac{3}{2} \quad m = -1$$

(ii) let roots be x and $\frac{1}{x}$.

then $x \times \frac{1}{x} = m + 3$

$$m + 3 = 1$$

$$m = -2$$

QUESTION 7

(a) $P = \left(\frac{kx_2 + lx_1}{k+l}, \frac{ky_2 + ly_1}{k+l} \right)$ where $k:l$ is $4:-3$ (x_1, y_1) is $(10, 2)$
and (x_2, y_2) is $(-2, 6)$.

$$= \left(\frac{4 \times -2 + -3 \times 10}{4-3}, \frac{4 \times 6 + -3 \times 2}{4-3} \right)$$

$$= (-38, 18).$$

(b) pt perp distance between $x-2y+k=0$ and $(0,0)$

$$d = \frac{|0 - 2 \times 0 + k|}{\sqrt{1 + (-2)^2}} = \sqrt{5}.$$

$$\frac{|k|}{\sqrt{5}} = \sqrt{5}$$

$$|k| = 5$$

$$k = \pm 5.$$

(c) Most likely solution by similar Δ s (could be done with intercept theorem).

(i) Show $\Delta ABC \parallel \Delta EFC$

$$\angle ABC = \angle EFC = 90^\circ \text{ (given)}$$

$$\angle ACB = \angle ECF \text{ (common)}$$

$\therefore \Delta ABC \parallel \Delta EFC$ (Equiangular).

$$\therefore \frac{h}{d} = \frac{b}{a+b} \text{ (1) (ratio of corresponding sides in congruent triangles)}$$

(ii) $\Delta DCB \parallel \Delta EFB$ (similarly to part (i)).

$$\therefore \frac{h}{c} = \frac{a}{a+b} \text{ (2)}$$

From (1) & (2)

$$\frac{h}{c} + \frac{h}{d} = \frac{a}{a+b} + \frac{b}{a+b}$$

$$h \left(\frac{1}{c} + \frac{1}{d} \right) = \frac{a+b}{a+b}.$$

$$\frac{1}{c} + \frac{1}{d} = \frac{1}{h} \text{ as required.}$$